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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION April 2026

Fourth SEMESTER

Multi-Disciplinary Courses (MDC) - MG2MDCMBG102 - MICROBES IN DAILY LIFE

(2024 ADMISSION ONWARDS)

Duration: 1 Hour **Maximum Marks: 35**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer any **10** questions

Each question carries **1** marks

1. What is the role of antioxidants in nutraceuticals? [K] / [CO1]

   Neutralise harmful free radicals to reduce oxidative stress, Preventing cellular damage and chronic diseases like cancer, diabetes, and cardiovascular issues. Protect against inflammation, support immune function, promote healthy aging, and enhance mitochondrial energy production.

2. Define the term "thermal death point". [K] / [CO3]

   The temperature necessary to kill all the organisms in 10 minutes

3. What is the relationship between probiotics and prebiotics? [U] / [CO1]

   Probiotics are live beneficial bacteria, and prebiotics are non-digestible fiber have a synergistic relationship where prebiotics act as food to nourish and promote the growth of probiotics in the gut.

   When used together, often called "synbiotics," they enhance digestive health, strengthen the gut barrier, and support immune function better than either alone.

4. What is a kiln? [K] / [CO3]

   Evaporator / Kiln – used in farm home – where the natural draft from the raising of heated air brings about the drying of food

5. Comment on Swiss cheese. [U] / [CO2]

   a mild, nut-like, and semi-hard cheese recognized by its signature "eyes" (holes) formed by *Propionibacterium* bacteria during fermentation. It is highly nutritious, offering significant amounts of protein, calcium, and Vitamin B12.

6. Give one example of food preserved by chemicals. [K] / [CO3]

Fruit juice, Processed Meats (Bacon/Sausages), Bread, Pickles/Jams.

7. Name a fermented cereal-based food. [U] / [CO1]

Idli, Dosa

8. What are the factors affecting the microbial flora of soil? [K] / [CO1]

Soil moisture, temperature, pH, aeration, and organic matter content.

9. What is frozen storage? [K] / [CO3]

The product is stored in a frozen state, preserving perishable items, long term storage.

10. What is Resident microflora? [U] / [CO1]

constant, cannot be removed permanently, Prevent permanent colonization by pathogens.

11. Define Infection dust. [K] / [CO1] microscopic particles of dust, soil, or dried organic matter carrying bacteria, viruses, or fungi, which cause disease when inhaled.

12. Mention the importance of microorganisms in food industry. [K] / [CO1]

for fermentation (yogurt, cheese, bread, beer), extending shelf life through biopreservation, improving nutritional value (probiotics), and producing food additives like vitamins, enzymes, and colors.

[1x10 = 10]

Part B

Short Essay Questions

Answer any **3** questions

Each question carries **5** marks

13. Discuss the production of wine [K] / [CO2]

Introduction, distinct stages: crushing, fermentation, aging, and bottling.

14. Write a short note on chemical methods of food preservation. [U] / [CO3]

Chemical preservatives – definition with examples, food additives.

Use of Preservatives, Mechanism of Action

15. Write a short note on the nutritional value of curd. [K] / [CO2]

Curd – definition, microorganisms involved in the production, Nutritional composition - High-Quality Proteins, Essential Vitamins and Minerals (Calcium and Phosphorus, Vitamin B complex, Vitamin A), Probiotic Benefits, Digestibility and Lactose Intolerance.

16. Describe the mechanism of action of sodium benzoate as a preservative. [U] / [CO3]

Preservation – depends on the pH of the food (pH 2.5 to 4.0). used in fruit juices, carbonated drinks, and pickles. mechanism of action - Conversion to Benzoic Acid, Intracellular pH Disruption, Metabolic Exhaustion, Enzyme Inhibition.

17. Discuss the impact of fermented foods on the immune system. [U] / [CO1]

Strengthening the Gut Barrier, Modulation of Immune Cells, Production of Short-Chain Fatty Acids (SCFAs), Competitive Exclusion, Influence on the Gut-BrainImmune Axis

18. Explain the concept of natural preservatives with examples. [U] / [CO3]

[5x3 = 15]

natural preservatives – definition (derived from nature—plants, animals, or minerals with examples). Mechanism of Action - Antimicrobial Activity, Antioxidant Activity, pH Alteration. Advantages and disadvantages.

Part C

Long Essay Questions

Answer any **1** questions

Each question carries **10** marks

19. Compare the fermentation processes in coffee and tea production. [U] / [CO2]

Introduction – coffee and tea (definition), comparison, production – detail, specific enzymes and microorganisms involved.

20. Discuss physical, chemical, and biological methods of food preservation with examples. [An] / [CO3]

Introduction – preservation, physical Methods -Thermal Processing (Heat), Low-Temperature Preservation, Dehydration (Drying), Irradiation; Chemical Methods - Traditional Preservatives, Chemical Additives, Gaseous Preservatives; Biological Methods (Biopreservation) – Fermentation, Bacteriocins, Protective Cultures.

[10x1 = 10]